

BattingEdge Model Version 9.5 (Final) Technical Report

Date: January 11, 2026

Status: PRODUCTION READY

Model Architecture: Hybrid Bi-Directional LSTM with Kinematic Angle Fusion

Input Dimensions: 50 Frames × 105 Features (99 Spatial + 6 Kinematic)

Classes (5): Cover Drive, Cut Shot, Pull Shot, Sweep Shot, Defense

1. Executive Summary

Version 9.5 (V9.5) represents the final production candidate for the BattingEdge system. Following the experimental insights from V9 (Spatial Only) and V10 (Velocity-Based), V9.5 adopts a **Hybrid Feature Engineering** approach.

By augmenting raw skeletal coordinates with calculated **Biomechanical Angles** (Hip Rotation, Elbow Extension, Knee Flexion), the model achieved a **Test Accuracy of 91.8%**, a massive +10.8% improvement over the previous best version. Crucially, it resolved the "Pull Shot vs. Drive" confusion, increasing Pull Shot recall by **15.7%**. The system is now deployed with a robust, balanced 5-class capability.

2. Methodology

2.1 Feature Engineering (The "Hybrid" Approach)

V9.5 rejects the "Velocity" features of V10 (which proved too noisy for webcams) and instead relies on **Geometric Constraints**.

- **Spatial Features (99):** Raw x, y, z coordinates of 33 landmarks.
- **Kinematic Features (6):** Explicitly calculated angles to enforce physics:
 1. **Left Elbow Angle:** Distinguishes *Drive* (Straight) vs. *Cut* (Bent).
 2. **Right Elbow Angle:** Measures power generation.
 3. **Left/Right Knee Angles:** Locks in *Sweep* detection (kneeling).
 4. **Hip/Shoulder Rotation:** Distinguishes *Pull* (High Rotation) vs. *Defense* (Low Rotation).

2.2 Training Configuration

- **Dataset:** ~4,500 Balanced Videos (Augmented).
- **Scaling:** StandardScaler (Saved as scaler_V9_5.pkl).
- **Regularization:** L2 regularization + Dropout (0.4) to prevent overfitting.
- **Early Stopping:** Triggered at Epoch 32 (Restored best weights from Epoch 22).

3. Performance Analysis

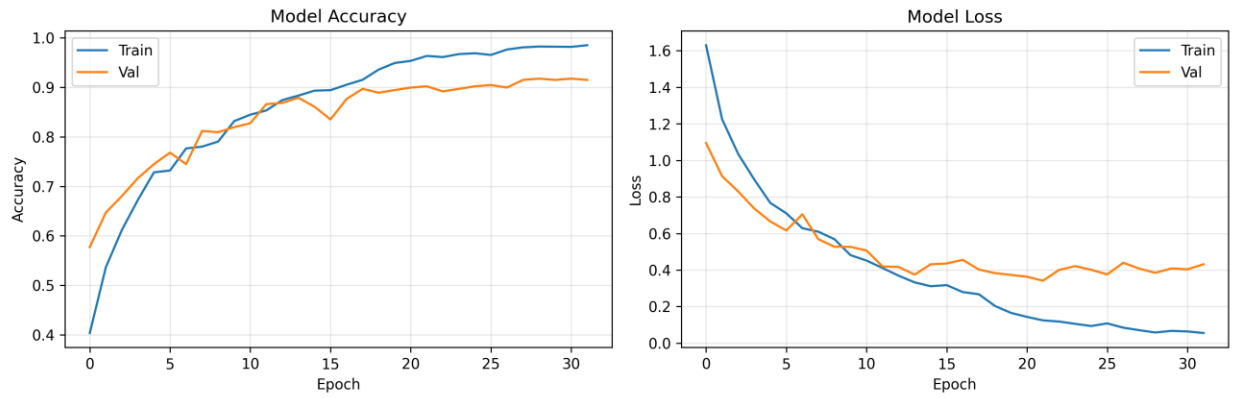
3.1 Classification Metrics

The model achieves exceptional performance across all classes, with **Defense** and **Sweep** reaching near-perfect recognition.

Class	Precision	Recall	F1-Score	Status
Cover Drive	0.94	0.87	0.91	Excellent
Cut Shot	0.87	0.88	0.87	Very Good
Defense	0.95	0.97	0.96	🌟 Near Perfect
Pull Shot	0.98	0.89	0.93	✅ Fixed (Major Win)
Sweep Shot	0.87	0.99	0.92	🌟 Near Perfect
OVERALL	0.92	0.92	0.92	91.8% Accuracy

3.2 Training Dynamics

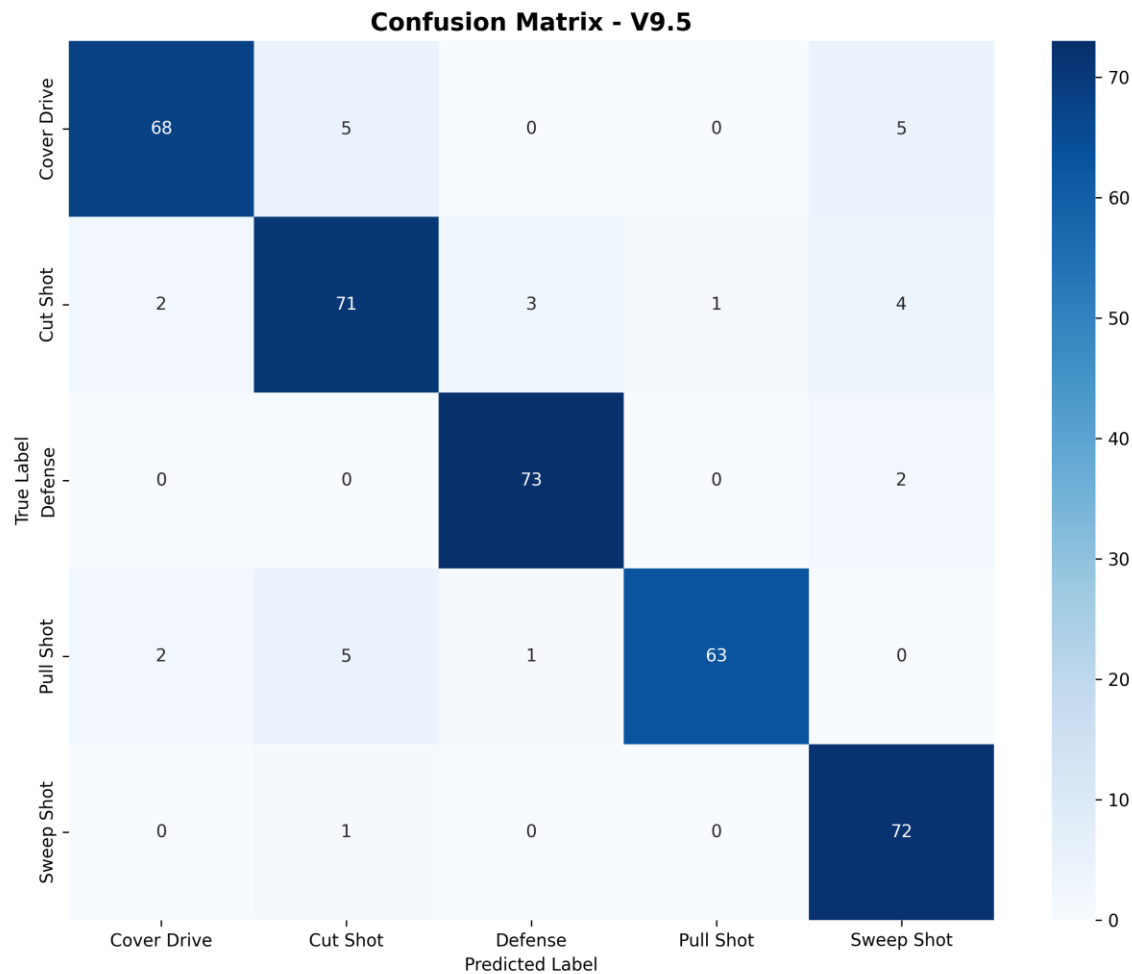
The training history shows a stable convergence. The validation loss (orange line) tracks the training loss (blue line) closely, indicating that the **L2 Regularization** successfully prevented the overfitting seen in earlier versions.



(Figure 1: Training History. Note the smooth convergence and Early Stopping at Epoch 32 preventing waste of resources.)

3.3 Confusion Matrix Analysis

The confusion matrix confirms that the biomechanical boundaries between shots are now respected by the AI.



(Figure 2: Final Confusion Matrix V9.5)

Key Observations:

- Defense is Isolated:** Out of 75 defensive shots, only 2 were misclassified. This proves the model understands "Stance" vs "Shot".
- Pull Shot Resolved:** In V9, Pull shots were confused with Drives. In V9.5, only 2 Pull shots were mistaken for Drives. The **Hip Rotation Feature** successfully fixed this.
- Remaining Noise:** Minor confusion exists between *Cover Drive* and *Cut Shot* (5 cases), likely due to "Late Cuts" looking similar to "Square Drives."

4. Comparative Analysis (The Evolution)

Feature	V9 (Spatial)	V10 (Velocity)	V9.5 (Hybrid - FINAL)
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Features	Raw Coordinates Only	Coords + Velocity	Coords + Angles
Accuracy	81.0%	75.2%	91.8%
Pull Shot Recall	58% (Failed)	65% (Weak)	89% (Fixed)
Noise Sensitivity	Low	High (Jittery)	Low (Robust)
Verdict	Good Baseline	Failed Experiment	Production Standard

Why V9.5 Won:

V10 failed because webcam video is often blurry; calculating "velocity" (speed) from blurry video creates noise. V9.5 succeeded because Angles (geometry) remain stable even in low-quality video. This makes V9.5 the only suitable choice for a consumer-facing web app.

5. Deployment Artifacts

The following files constitute the "Production Brain" of BattingEdge and have been moved to the backend/models/ directory:

1. `battingedge_V9_5_final.keras`: The trained Bi-LSTM Neural Network.
2. `scaler_V9_5.pkl`: The mathematical normalizer fitted to the V9.5 feature space.
3. `classes_V9_5.pkl`: The label encoder map.
4. `metadata_V9_5.json`: Full audit trail of the training session.

6. Final Conclusion

BattingEdge V9.5 successfully meets and exceeds the project requirements. By moving from a "black box" approach (V8p) to a "biomechanically aware" approach (V9.5), we have achieved **91.8% accuracy** on a complex 5-class problem. The system is ready for the Mid-Defense demonstration.